CALMCARE

33 features — explained accurately and clearly

Detailed feature specification for the H2 hackathon problem: Discharge Summary to Patient-Understandable Care Plan

This revised document explains what each feature does, how the website should behave, an example, and the safety boundary that must be respected.

Our central promise

Make discharge instructions easier to understand without changing what the doctor actually wrote. CalmCare must make uncertainty visible and must never invent a medicine dose, timing, appointment or treatment instruction.

This is a revised feature specification, not a claim that all 33 features are required by the hackathon or that no similar feature exists in any other product.

How to read this document

Each feature is described in four parts: **Purpose** (the problem it solves), **Website behaviour** (what happens when someone uses it), **Example** (a concrete user interaction), and **Safety boundary** (what the system must not do). This structure is intended to make implementation and judging clearer.

The official H2 problem statement specifically prioritises faithfulness over fluency. It asks for a structured care plan, English plus one Indian language, a readability score, a one-page fridge sheet, and optionally spoken audio and a 7-day medicine timetable. Multiple languages, camera capture, reminders, caregiver tools and the other advanced features are proposed extensions.

Suggested implementation order

Priority	Features / capability	Why
Core first	1–5, 8, 12	Accept the source, extract it, explain it, preserve facts and produce required outputs.
Safety next	2, 8, 15, 19, 29, 32, 33	Detect uncertainty, trace facts and avoid unsafe confident output.
Voice and access	6, 7, 9, 26	Make the verified plan understandable in text and supported spoken languages.
Daily support	10, 17, 18, 21, 27, 31	Only after schedules and changes are reliably verified.
Advanced support	11, 13, 14, 16, 20, 22–25, 28, 30	Improve usability, continuity and privacy once the core pipeline works.

1. Discharge-summary upload and camera capture

PURPOSE

Give patients two equally visible ways to provide the document: choose a file or photograph the paper summary directly in the website.

HOW IT WORKS ON THE WEBSITE

The user selects a PDF, JPG or PNG, or opens the camera. For paper documents, the user captures each page, previews it, rotates/crops it, retakes unclear images, reorders pages and confirms that all pages are included. The site shows the selected files before processing.

EXAMPLE

A patient has a two-page paper discharge summary. They photograph page 1 and page 2, check that neither page is cut off, then submit both together.

SAFETY BOUNDARY

Do not begin generating a care plan from a visibly incomplete document. Protect camera and file permissions; camera use must be initiated by the user.

2. Smart medical information extraction

PURPOSE

Turn the document into structured fields that can be checked, rather than asking the language model to rewrite a large block of text without a record of its facts.

HOW IT WORKS ON THE WEBSITE

Read digital PDF text directly where possible; run OCR on scanned PDFs and photos. Extract medicine name, dose, unit, route if stated, frequency, duration, follow-up date, warning signs and diet/activity restrictions. Store the source passage/page for each field and mark confidence or uncertainty.

EXAMPLE

The system extracts “Medicine A — 1 tablet — twice daily — 5 days” and keeps the page/line where this appears so a reviewer can compare it.

SAFETY BOUNDARY

OCR is not always correct, especially for decimal points, units and handwritten text. Uncertain critical fields must be flagged for review rather than silently corrected.

3. Plain-language illness and discharge explanation

PURPOSE

Explain what the discharge summary says in words a person with little or no medical knowledge can understand, aiming for roughly Class 8 reading difficulty where measurable.

HOW IT WORKS ON THE WEBSITE

Explain medical terms using short sentences, familiar words and small sections. Separate what the document says about the condition from the actions the patient is instructed to take. Explain the purpose of a treatment only if that purpose is supported by the source or approved reference content.

EXAMPLE

Instead of only showing “hypertension”, explain it as “high blood pressure”, while preserving the medical term so the patient can recognise it on the document.

SAFETY BOUNDARY

Do not add a diagnosis, cause, prognosis, treatment benefit or promise of recovery that the source does not support. Keep exact prescribed values visible.

4. Structured patient care plan

PURPOSE

Organise instructions into consistent sections so the patient can find what to do without reading the entire hospital document.

HOW IT WORKS ON THE WEBSITE

Create sections for medicines; warning signs and when to seek help; diet; activity/rest or wound care if documented; and follow-up appointments. Each item has a plain-language explanation and source reference. Missing information is labelled “not stated in this summary,” not filled in.

EXAMPLE

A patient sees separate cards for “Medicines”, “When to get help”, and “Next appointment”, rather than one long paragraph.

SAFETY BOUNDARY

Only include source-supported or explicitly approved material. Do not invent diet restrictions, exercises, wound-care steps or emergency symptoms.

5. Multiple selectable languages

PURPOSE

Let each patient choose languages they can read or understand instead of limiting the product to a single fixed regional language.

HOW IT WORKS ON THE WEBSITE

Offer English and a list of Indian languages that the team has actually integrated and tested. The patient may select one or more output languages and switch between them without uploading again. Keep the underlying verified facts shared across versions.

EXAMPLE

A patient reads the plan in Telugu and chooses Hindi for a caregiver, while both versions use the same validated dose and date.

SAFETY BOUNDARY

Do not claim support for a language until translation and, if offered, speech output have been tested. Validate numbers, units, dates, frequencies and warnings separately in every language.

6. Voice and text interaction

PURPOSE

Let patients who find typing difficult speak to the website, while keeping a visible text alternative for clarity and accessibility.

HOW IT WORKS ON THE WEBSITE

The patient taps the microphone, speaks a request such as “Explain my next appointment”, and sees the speech-to-text transcript. They can edit or confirm it before the system acts. A text box remains available if microphone permission or speech recognition fails.

EXAMPLE

The user says “Speak in Telugu”. The website displays the recognised request and asks for confirmation if the transcript is uncertain.

SAFETY BOUNDARY

Speech recognition can mishear medicine names, numbers and dates. Never rely on uncertain voice input to alter a prescription or schedule.

7. CalmCare spoken guidance

PURPOSE

Make the plan usable for patients who prefer listening or cannot comfortably read a long document.

HOW IT WORKS ON THE WEBSITE

Read one topic at a time in a calm, warm and patient voice. Start slowly, pause after important instructions, and provide pause, replay, slower/faster playback and section-by-section controls. Display synchronized text wherever possible.

EXAMPLE

The voice explains the medicine section, pauses, and asks whether the patient wants it repeated before moving to follow-up instructions.

SAFETY BOUNDARY

Voice quality and pronunciation vary by language and provider. Test actual supported voices. Do not use a soothing tone to hide uncertainty or give false reassurance.

8. Medical-instruction verification engine

PURPOSE

Check that the generated care plan has not dropped, changed or invented critical medical details.

HOW IT WORKS ON THE WEBSITE

Compare the structured source facts with the output: medicine identity, dose, unit, frequency, duration and follow-up date. Check for missing facts, changed values and unsupported additions. Produce a list of mismatches and block “verified” status while a critical issue is unresolved.

EXAMPLE

If the source says “twice daily” but the output says “once daily”, the checker reports a frequency mismatch and prevents that plan from being labelled verified.

SAFETY BOUNDARY

A passed check is evidence of consistency with the source, not proof that the source itself is clinically correct. Do not claim perfect medical safety.

9. Low-literacy and accessibility mode

PURPOSE

Make the website usable for people who struggle with small text, dense paragraphs, low vision, reading or navigation.

HOW IT WORKS ON THE WEBSITE

Offer large text, high contrast, clear labels, short sections, familiar icons, keyboard navigation and screen-reader-friendly controls. Pair important icons with words and provide audio alternatives.

EXAMPLE

A patient turns on large text and hears the same medicine instructions read aloud while the exact dose remains visible.

SAFETY BOUNDARY

Icons must not replace important wording. Accessibility settings must not hide warnings, units or other critical details.

10. Visual medicine timetable

PURPOSE

Turn a verified prescription into a clear visual schedule when the source contains enough timing detail.

HOW IT WORKS ON THE WEBSITE

Display each medicine with its prescribed dose, frequency and duration. Use exact time slots only when specified or confirmed through an authorised review workflow. If only a frequency is given, show that frequency without inventing clock times.

EXAMPLE

If a summary specifies morning and evening, show those slots. If it only says “twice daily”, do not silently assign 8 AM and 8 PM.

SAFETY BOUNDARY

Never change a dose or recommend when to take it beyond the source. Unclear timing must remain unresolved until confirmed.

11. Patient and caregiver mode

PURPOSE

Allow a trusted family member to help the patient understand the plan without removing the patient's control over their information.

HOW IT WORKS ON THE WEBSITE

The patient chooses whether to share the plan and with whom. A caregiver can view the same verified instructions and, if permitted, help with reminders or appointments. Clearly distinguish patient-confirmed actions from caregiver-reported actions.

EXAMPLE

A patient gives their daughter permission to view the Telugu plan and receive appointment reminders.

SAFETY BOUNDARY

Sharing is opt-in and revocable. Do not assume that a family member is authorised to access the patient's health information.

12. One-page fridge sheet and quality report

PURPOSE

Provide a practical printed summary and a separate explanation of how well the digital plan passed checks.

HOW IT WORKS ON THE WEBSITE

Generate a one-page sheet with verified key medicine instructions, documented warning signs and follow-up details. Separately show readability, missing fields, unresolved discrepancies, unsupported additions and per-language validation results.

EXAMPLE

The family prints the key plan for the refrigerator and can separately inspect a report showing that a follow-up date needs review.

SAFETY BOUNDARY

Do not squeeze the page so tightly that doses or warnings become unreadable. Never hide unresolved safety problems to make the report look successful.

13. Ask My Discharge Summary assistant

PURPOSE

Answer patient questions about the uploaded plan without acting as an unrestricted medical chatbot.

HOW IT WORKS ON THE WEBSITE

Accept typed or spoken questions and retrieve the relevant verified source fact or approved reference. Provide a concise answer with a link to the source passage. If the summary does not answer the question, say that clearly and suggest contacting the care team.

EXAMPLE

Asked “When is my follow-up?”, it repeats the documented date. Asked “Should I double a missed dose?”, it does not guess and directs the patient to their clinician/pharmacist or official instructions.

SAFETY BOUNDARY

Answers must be grounded. The assistant must not diagnose, prescribe, infer missing directions or delay urgent care.

14. Teach-back understanding check

PURPOSE

Help identify which instructions may not yet be clear, without testing or shaming the patient.

HOW IT WORKS ON THE WEBSITE

Offer optional simple questions based on the verified plan, such as selecting the documented appointment date. If the answer is incorrect or the patient says they do not know, calmly re-explain the same source-supported instruction and offer voice or visual support.

EXAMPLE

The patient chooses the wrong follow-up date. CalmCare repeats the date slowly and shows the original source sentence.

SAFETY BOUNDARY

This is not a diagnosis or a measure of intelligence. Do not infer non-adherence or clinical risk from one incorrect answer.

15. Uncertainty and contradiction detection

PURPOSE

Make unclear source text visible instead of letting the system produce a confident but potentially wrong answer.

HOW IT WORKS ON THE WEBSITE

Flag blurry OCR, incomplete fields and contradictory medicine entries or dates. Show the relevant passages and request a clearer image, confirmation or authorised clinical review.

EXAMPLE

One page appears to say 5 mg and another appears to say 50 mg. CalmCare shows the conflict and does not select either value.

SAFETY BOUNDARY

Do not automatically choose the more common dose or the latest-looking entry unless a defined authorised review process confirms it.

16. Empathetic communication mode

PURPOSE

Make explanations respectful and emotionally considerate, especially when a patient is anxious, confused or asks the same question repeatedly.

HOW IT WORKS ON THE WEBSITE

Use neutral, kind language; acknowledge uncertainty; avoid blame; explain one action at a time; and allow the patient to pause or ask someone for help.

EXAMPLE

Instead of “You failed to follow the instructions”, say “Would it help if we went through this step together?”

SAFETY BOUNDARY

Empathy must not become false reassurance. Never promise a recovery outcome or minimise a warning sign.

17. Follow-up and appointment organiser

PURPOSE

Turn appointment details already written in the summary into a clear list and optional calendar entry.

HOW IT WORKS ON THE WEBSITE

Extract documented date, time, location and department if present. Show the details for confirmation before creating a calendar event. Allow export and user-requested rescheduling notes.

EXAMPLE

The summary contains a follow-up date but no time. The app records the date and marks the time as not stated rather than inventing one.

SAFETY BOUNDARY

Never invent appointment dates, locations or clinicians. Calendar creation and notifications require user permission.

18. Reminder reliability and missed-task handling

PURPOSE

Make reminders useful without assuming that silence means a medicine was missed.

HOW IT WORKS ON THE WEBSITE

Offer configurable snooze, quiet hours and reminder frequency. Allow the user to mark a dose taken, snooze it or record that it was not taken. Track only explicit user-confirmed actions.

EXAMPLE

If the user does not tap a reminder, the dashboard says “No response recorded”, not “Medicine missed”.

SAFETY BOUNDARY

Do not recommend doubling, skipping or changing a dose. Medication-specific missed-dose questions should be referred to the clinician/pharmacist or official instructions.

19. Clinical fidelity dashboard

PURPOSE

Show exactly which safety and quality checks passed or failed, rather than displaying one vague AI accuracy score.

HOW IT WORKS ON THE WEBSITE

Report medicine and date preservation, unresolved conflicts, unsupported additions, OCR uncertainty, readability and language-specific errors. Show examples of failures and test-set performance.

EXAMPLE

The report may show that the English version passed critical-field checks while one regional translation needs review.

SAFETY BOUNDARY

Never present a single aggregate score as proof of clinical safety. Separate measured results from assumptions and demo claims.

20. Low-bandwidth and print-friendly access

PURPOSE

Keep essential plan information accessible when the connection is weak or a patient prefers paper.

HOW IT WORKS ON THE WEBSITE

Use lightweight pages, avoid unnecessary animations, allow a verified plan to be downloaded/printed, and consider local viewing of a previously saved plan if the device supports secure storage.

EXAMPLE

A patient with limited connectivity opens a previously saved PDF of their verified plan.

SAFETY BOUNDARY

Offline copies can become outdated. Show the plan version/date and warn that a newer plan may supersede it. Protect local health information on shared devices.

21. Smart notifications and reminders

PURPOSE

Send optional prompts based on documented medicine timing, appointment dates and explicit care tasks.

HOW IT WORKS ON THE WEBSITE

Support in-app or browser/mobile push notifications where available, and optional SMS/messaging or voice reminders only with consent and an integrated provider. Provide taken/snooze/skip controls for medicine prompts and completion/reschedule controls for appointments.

EXAMPLE

A reminder repeats the medicine name and prescribed instruction already verified in the plan. A follow-up alert uses the date recorded in the summary.

SAFETY BOUNDARY

Only create schedules from verified details. Diet, hydration, exercise and wound-care prompts must come from documented or approved instructions; do not add generic advice that may conflict with restrictions.

22. Daily recovery check-in

PURPOSE

Let patients record how they feel and whether they completed tasks, so they can reflect on their experience and share it if they choose.

HOW IT WORKS ON THE WEBSITE

Ask brief optional questions about self-reported symptoms, concerns and documented task completion. Display trends over time and allow a user to share them with their care team.

EXAMPLE

The patient records “more tired today” for several days; the app displays the trend without diagnosing its cause.

SAFETY BOUNDARY

Do not diagnose from check-ins. If the user reports potentially urgent symptoms, show appropriate urgent-care guidance promptly rather than continuing a routine survey.

23. Personal recovery progress dashboard

PURPOSE

Give patients a simple view of their own recorded actions and check-ins after discharge.

HOW IT WORKS ON THE WEBSITE

Summarise confirmed medicine actions, appointment status, documented tasks and patient-reported trends. Label unknown or unconfirmed items clearly and allow correction of user-entered records.

EXAMPLE

The dashboard shows two appointments marked completed and three days of self-reported check-ins.

SAFETY BOUNDARY

A completed checklist does not prove clinical recovery. Do not create a recovery percentage or medical conclusion without validated clinical justification.

24. Warning-sign guidance

PURPOSE

Make warning signs already stated in the discharge summary easier to recognise and understand.

HOW IT WORKS ON THE WEBSITE

Display the documented signs, explain them in plain language and preserve the source's urgency wording. If a patient reports a potentially urgent symptom, direct them to appropriate urgent medical help immediately.

EXAMPLE

A warning sign listed in the summary is shown in a prominent section and can be read aloud in the chosen language.

SAFETY BOUNDARY

Do not invent warning signs or delay urgent guidance with lengthy questions. Do not replace local emergency services or clinician advice.

25. Condition-specific recovery guidance

PURPOSE

Organise the provided categories—diabetes, post-operative, cardiac, respiratory and maternity—so relevant documented sections are easy to find.

HOW IT WORKS ON THE WEBSITE

Use the condition category to choose the layout and approved glossary/reference content, while taking patient-specific instructions from the discharge summary. Keep generic educational content clearly separate from personal instructions.

EXAMPLE

A post-operative plan may place documented wound-care and activity restrictions in prominent sections if those items appear in the summary.

SAFETY BOUNDARY

Do not generate condition-specific advice merely because a condition label is detected. Additional clinical content must be reviewed and approved.

26. Explain this again

PURPOSE

Make it easy to ask for a second explanation of any individual instruction without restarting the entire plan.

HOW IT WORKS ON THE WEBSITE

Place controls next to each item: repeat, slower, simpler words, play audio or switch language. Keep the original prescribed value and source link accessible.

EXAMPLE

The patient taps “Explain again” beside a follow-up instruction and hears the same date in slower, simpler language.

SAFETY BOUNDARY

Rephrasing must not alter dose, timing, duration, warning level or other clinical meaning.

27. Discharge-instruction change management

PURPOSE

Prevent an older plan and its reminders from being silently used after a new discharge summary is uploaded.

HOW IT WORKS ON THE WEBSITE

Compare the new document with the prior verified plan. Show additions, removals and changes to medicine details, dates and tasks. Mark affected reminders for review; apply confirmed changes and retire obsolete reminders only after review.

EXAMPLE

A new summary lists a different medicine duration. CalmCare highlights the difference and pauses the old duration-based reminders pending confirmation.

SAFETY BOUNDARY

Document order alone does not prove which conflicting instruction is correct. Unclear changes require confirmation or clinical review.

28. Privacy-first processing

PURPOSE

Protect sensitive health information and give patients control over how it is used and shared.

HOW IT WORKS ON THE WEBSITE

Collect only necessary data, use secure transfer/storage, avoid health details in logs, explain retention, provide deletion controls and request consent before caregiver sharing or optional messaging. Design the prototype to work from the discharge summary without full EHR access.

EXAMPLE

A user can delete an uploaded document and its saved plan using a clearly labelled control, subject to the system's disclosed retention limits.

SAFETY BOUNDARY

Do not promise deletion or security properties the implementation cannot actually deliver. Limit access and test the deletion flow.

29. Source-proof for every medical instruction

PURPOSE

Let a patient or reviewer check where an important instruction came from.

HOW IT WORKS ON THE WEBSITE

Attach a source passage and page location to every critical structured fact. A user taps a medicine or date to view the supporting sentence in the original document.

EXAMPLE

Tapping the dose opens the page region containing the dose in the discharge summary.

SAFETY BOUNDARY

If no reliable source span is available, label the item unverified instead of fabricating a citation or implying that the source supports it.

30. Adaptive understanding engine

PURPOSE

Choose a more helpful explanation style based on the patient's expressed needs, not change the treatment instructions.

HOW IT WORKS ON THE WEBSITE

Use optional teach-back responses, “I don't understand” feedback and repeat requests to offer a simpler version, slower audio, visual schedule or translation. The patient can choose a preferred style.

EXAMPLE

After a patient says a term is confusing, the app explains the same source-supported instruction with simpler words and plays it slowly.

SAFETY BOUNDARY

Do not profile or label a patient as incapable. Do not alter medical meaning or infer clinical adherence from interaction behaviour.

31. Discharge change impact analyser

PURPOSE

Go beyond listing text differences by identifying which parts of the user's active plan may be affected by a new document.

HOW IT WORKS ON THE WEBSITE

Map changed source facts to dependent outputs: medicine timetable, reminders, fridge sheet, calendar entries and caregiver view. Present a change summary and require confirmation/review before replacing active items.

EXAMPLE

If the documented follow-up date changes, the app flags the old appointment reminder and proposes updating it after confirmation.

SAFETY BOUNDARY

Never silently update a critical schedule when the new document is unclear or contradictory. Keep an audit trail of the confirmed plan version.

32. Cross-language meaning consistency checker

PURPOSE

Check whether each translated care plan preserves the verified medical facts—not just whether the translation reads naturally.

HOW IT WORKS ON THE WEBSITE

Compare medicine names, numerals, units, frequency, duration, dates, warnings and restrictions between the verified structured record and every language output. Run language-aware checks and flag differences for correction/review.

EXAMPLE

If “twice daily” is accidentally translated as “once daily”, the affected language version is marked as needing review and cannot be labelled verified.

SAFETY BOUNDARY

Translation checks are imperfect. A passed automated check is not proof of perfect meaning; critical language output needs testing and, where possible, qualified review.

33. Patient-specific instruction conflict detector

PURPOSE

Identify contradictory instructions inside a single discharge summary before they become a confident patient-facing plan.

HOW IT WORKS ON THE WEBSITE

Compare repeated medicine entries, doses, frequencies, durations and appointment dates. Show the conflicting passages side by side and mark the affected field as unresolved.

EXAMPLE

A medicine appears as 5 mg in one section and 50 mg in another. CalmCare highlights both passages and requests confirmation rather than choosing one.

SAFETY BOUNDARY

Do not resolve a conflict by guessing from common practice, drug references or model confidence. Require authorised confirmation or clinical review.

Implementation rules that connect all 33 features

These rules should be implemented centrally so that features do not contradict one another.

One verified source record

Store each extracted fact with its source text/page, value, units, confidence and review status. The care plan, translations, reminders and printouts should be generated from this shared record.

No verified record, no active schedule

If a critical medicine detail or time is unclear, do not activate the affected reminder or present an exact timetable until it is resolved.

Separate source facts from explanations

Keep the exact clinical fact distinct from its simplified explanation. Rewriting the explanation must never mutate the medicine or date fields.

Translations are separate outputs, not separate medical truth

Every language version must be checked against the same verified fact record. A language version with a critical mismatch is flagged even if another language passed.

Changes require review

When a new discharge summary arrives, compare versions, identify affected reminders and ask for confirmation or authorised review before applying ambiguous changes.

Show uncertainty plainly

Use understandable labels such as “Please check this dose” or “The time is not written here”. Do not hide uncertainty in a technical log only.

Consent for sharing and notifications

Caregiver sharing, push notifications and optional messaging must be user-controlled and revocable.

Measure rather than exaggerate

Report actual test results for omissions, altered values, unsupported additions, language fidelity, readability and usability. Do not claim clinical outcomes without evidence.

Final definition

CalmCare is a multilingual, voice-first, safety-verified recovery companion. It turns a discharge summary into a plain-language care plan, explains it patiently, connects critical instructions to their source, flags uncertainty, and helps patients organise documented tasks without inventing medical advice.